

Ashmit Dutta

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EDUCATION

University of Illinois Urbana-Champaign

BSc, Computer Engineering

GPA: 3.5/4.0

Aug 2023 – May 2027

EXPERIENCE

Assured and Trustworthy AI Research Lab, UIUC

Urbana, IL

Undergraduate Research Assistant (Prof. Huan Zhang)

Dec 2025 – Present

- Contributing features in Python and CUDA to **auto-LiRPA**, the core verification library behind α, β -CROWN, winner of VNN-COMP 2023–2025.
- Engineering evaluation pipelines and benchmark datasets for the lab's VNN-COMP 2026 global submission.

John Deere

Champaign, IL

Software Engineer, Part-Time Student

Jan 2025 – Jan 2026

- Modernized exporting platform to production for 500+ users, migrating to React, Spring Boot, and AWS; designed core API integrating PostgreSQL and DB2 for millions of records.
- Integrated AWS CloudWatch for real-time observability, reducing incident resolution time (MTTR) by 40%.
- Engineered automated CI/CD pipelines to cut deployment time 50% and secured sensitive export data on AWS S3.

The University of Chicago Data Science Institute

Chicago, IL

Machine Learning Research Intern

Jun 2024 – Aug 2024

- Built a multi-modal GNN library in PyTorch Geometric for Fermilab's Exa.TrkX neutrino reconstruction project, improving model accuracy to 98.3% with stacked message-passing layers.
- Processed 20,000+ neutrino interaction events with H5Py on UChicago's HPC cluster (Slurm), enabling end-to-end training on real detector data.

PROJECTS

Autonomous Drone Racing (Crazyflie) | ROS, Python, PyTorch, CasADi, CRTP

Oct 2025 – Dec 2025

- Built a full ROS autonomy stack for a Crazyflie quadrotor: geometric SE(3) controller, CasADi minimum-snap trajectory planner, and Python hardware interface bridging planners to embedded firmware over CRTP radio.
- Finetuned a YOLOv8-seg gate segmentation model in PyTorch on NeRF-rendered synthetic data using a multi-objective Bayesian hyperparameter optimizer; **fastest drone in ECE 484 (Safe Autonomy) class**.

ShotSpot: Multi-Modal Video Search Engine | Python (Modal, BrightData, FastAPI), MongoDB

Feb 2026

- Winner (Modal Inference Track + Bright Data AI) at Stanford TreeHacks 2026**; built a distributed full-stack video search and dataset generation app using cloud GPU inference.
- Designed a multi-modal embedding pipeline fusing vision, audio, and text for semantic retrieval; integrated Bright Data scraping agents for automated ingestion, deployed on Modal with MongoDB.

GPU Accelerated Deep Learning Kernel (LeNet-5) | C++, CUDA

Feb 2026 – Apr 2026

- Developed high-performance 2D convolution kernels for LeNet-5 inference on CPU (C++) and CUDA GPUs, converting convolution to GEMM via im2col and applying kernel fusion to reduce memory traffic.
- Profiled with Gprof and NVIDIA Nsight, then implemented shared-memory tiling, CUDA streams, and Tensor Core acceleration for sub-10 ms inference. **Placed 1/130 students in ECE 408 class competition for fastest inference kernel**.

RISC-V Operating System | C, Assembly, QEMU

Feb 2025 – May 2025

- Designed a Unix-like OS from scratch in C for RISC-V (paging, multitasking, file I/O, shell, QEMU drivers for UART/VIRTIO, syscalls, ELF loading, Sv39 virtual memory, fork/exec, pipes, preemptive scheduler); built a bare-metal hypervisor to run multiple of these OSes concurrently.

Out-of-Order Superscalar RISC-V Processor | SystemVerilog, Design Compiler, RVFI

Jan 2026 – Apr 2026

- Architected a 2-wide superscalar RV32IM processor in SystemVerilog with a merged 48-entry physical register file and 32-entry Reorder Buffer (ROB) to fetch, issue, and commit two instructions per cycle.
- Designed a high-throughput memory subsystem (2-slot 256-bit instruction linebuffer, split LSQ, dual-client cache arbiter, 8-entry store buffer) and tournament branch predictor (bimodal + GShare, 128-entry BTB, 8-entry RAS); **achieved 0.85 CoreMark IPC at 500 MHz**.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript, SystemVerilog, SQL, Assembly (RISC-V)

Frameworks: React, Spring Boot, PyTorch, CUDA, ROS, CasADi, Flask, Pandas, NumPy

Tools: Docker, AWS, Linux/Unix, STM32, PCB, Gazebo, GDB, Git, LaTeX, Slurm, QEMU, Wireshark